

USYD SoC Design Challenge – Agriculture SoC

Timeline and verification plan to tapeout 7 December 2026

Prepared by Lewis Watts for the team, 21 September 2026 · github.com/lewisMW/agriculture-SoC

Summary

We are taping out a mixed-signal precision-agriculture SoC using an Arm Cortex-M0 subsystem on the SoCLabs NanoSoC platform with an 8-bit, 100 kS/s SAR ADC front end on the ChipFoundry chipIgnite ci2612 shuttle (SkyWater SKY130). The project began in ELEC3607 (2024 S1) and competes in the SoCLabs SoC Design Contest (Educational Implementation track).

Project Status: the full digital chip is synthesised, placed, routed and timing-closed at 33 MHz; the ADC is verified at transistor level from firmware running on the Cortex-M0 (Fig. 1); every analog cell has a schematic and layout, DRC is essentially clean and LVS is under way. **Funding:** S3B provides AU\$21,000 against a US\$14,500 shuttle. **Key dates:** 50 % commit **8 October** (flexible), tapeout **7 December** (fixed).

1 Status and Recent Achievements

- **Digital implementation (Daniel Geng).** Cadence Genus synthesis and Innovus place-and-route of the SoC: Cortex-M0, four 8 kB SRAMs, SKY130 pad ring (Fig. 2). **Timing closed post-route at 33 MHz** and closing DRC violations.
- **Mixed-signal co-verification (Ben Schwarz & Lewis Watts).** C firmware on the Cortex-M0 triggers SAR ADC conversions over the ARM core's APB bus. The analog components use SKY130 transistor-level cells solved by Cadence Spectre running within an Xcelium AMS simulation. Six self-checking firmware tests (validated on a deliberately broken design) produce ADC codes within 1 to 2 LSB of the behavioural model.
- **Analog cells in SKY130 (Hee Chan Kim Cabrera, Erick Gomez Landeros Acosta, Jaime Pryor).** All eight analog modules have AKY130-specific schematics and layouts and are DRC-clean, apart from density rules that will be resolved by chip fill in the integration stage. LVS has started, 50% progress.
- **Independent reviewers:** Kushal Das (Senior Cryogenic ASIC Engineer, USyd Nano Institute): analog, already mentoring the SAR design; Daniel Newbrook (Senior ASIC/FPGA Design Engineer, University of Southampton / SoCLabs): SKY130 flow and Cortex-M0 integration; Samuel Tensingh: SKY130 Cadence flow.

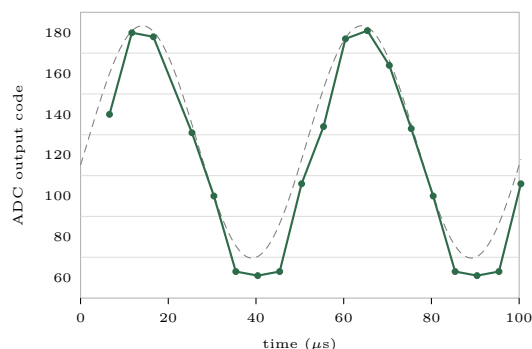
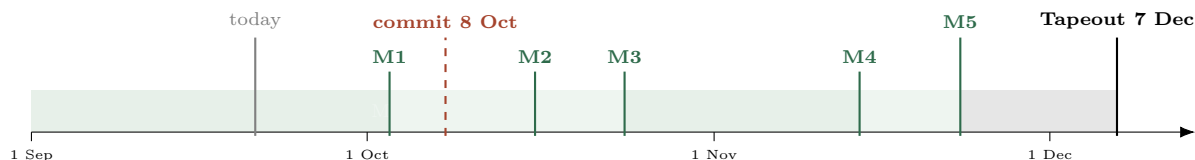


Fig. 1. Codes read by the Cortex-M0 from the transistor-level SAR ADC (Spectre in the loop) for a 20 kHz differential sine (dashed: ideal). The sine's troughs bias towards 0x3F, which is consistent with a known switch-recharge effect at twice the rated conversion rate, to be fixed.

Budget (AU\$, US\$ at 1.4035 mid on 21 Sep)

chipIgnite ci2612 slot, incl. eval PCB (US\$14,500)	20,350
50 % commit, 8 Oct (US\$7,250)	10,175
balance at tapeout, 7 Dec (US\$7,250)	10,175
FX transfer fees (est. 3.0 %)	610
Connectors, consumables (balance)	200
Total cost	21,000
S3B funding	–21,000
Balance (AU\$)	160

2 Timeline: Milestones to Tapeout



- M1 3 Oct** LVS passes on all analog modules; ADC macro layout + parasitic extraction, the simulation of which still passes the 6 self-checking firmware tests in mixed-signal co-verification.
- M2 10 Oct** The MVP of the ADC macro is provided for integration into the synthesis flow.
- M3 24 Oct** Achieve ADC figure of merits including: INL/DNL < 1 LSB, no missing codes, ENOB ≥ 7 at 100 kS/s across process/temperature/supply corners, and a process-based Monte-Carlo simulation produces voltage offsets within the configurable trim range.
- M4 14 Nov** Full-chip DRC/LVS/antenna clean; firmware regression tests passes with the full extracted ADC in the loop. Gate level simulation passes.
- M5 23 Nov** Sign-off checklist reviewed with Kushal Das, Daniel Newbrook and Samuel Tensingh with two weeks of buffer. GDS submitted 7 Dec.

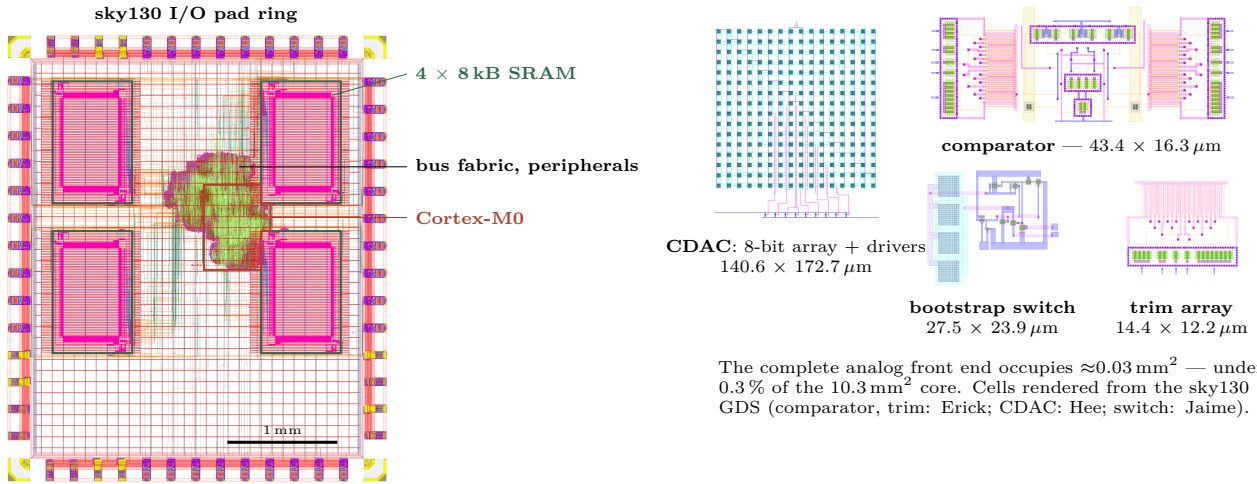


Fig. 2. Left: the routed SoC as of 21 Sep, post-route with filler, pre-signoff (filler cells and row rails hidden for clarity): $3.44 \times 4.04 \text{ mm}$ die, SKY130 pad ring, four 8kB SRAMs, Cortex-M0 (red), fabric logic. Right: four analog cells, enlarged.

3 Milestone 1 Status

Cell	Owner	DRC	LVS	Open item
inverter	Hee	clean	match	—
cap array (carray_8b)	Hee	clean	mismatch	schematic/layout unit-cap count to reconcile (127 vs 128)
CDAC (cdac_8b)	Hee	clean	to run	—
bootstrap switch	Jaime	clean	mismatch	recharge under rapid triggering to characterise
comparator	Erick	clean	mismatch	layout pin labels
trim array	Erick	clean	to run	ideal caps to be replaced by SKY130 MIM

6 of 6 cells DRC-clean (exc. density rules, to be resolved by chip fill in integration). 1 of 6 LVS-matched. Firmware regression: 6/6.

4 Verification Plan

Level	Physical	Electrical	System
ADC macro	DRC/LVS clean, parasitics extracted	ADC FOM simulated: INL/DNL, ENOB across corners	Extracted netlist passes the AMS test-bench regression tests
Chip	Full-chip DRC, LVS, ERC, antenna, fill	Static timing at 33 MHz post-route (done); review with mentors	All firmware tests pass in AMS + gate level simulation

5 Risks and Mitigation Plan

Risk	Mitigation
ADC misses spec	Digital SoC is already timing-closed and firmware-verified: silicon returns as a working Cortex-M0 MCU regardless and the ADC is characterisable via the firmware.
ADC switch recharge limits rate	Increase the minimum conversion interval in firmware.
Comparator offset exceeds trim	Cap sizing reviewed by Kushal Das in M1, before the layout is frozen.
Schedule slip	Two weeks of buffer time before 7 Dec.

6 Physical Test Plan

ChipFoundry's chipIgnite delivers 100 QFN parts, an evaluation board and 10 parts on M.2 cards five months after tapeout (~May 2027). Bring-up procedure: (1) boot the Cortex-M0 over SWD with the regression firmware images; (2) characterise the ADC from a signal generator through the same firmware sensing tests and measure key FOM (INL/DNL from a slow ramp, ENOB from an FFT of captured codes); (3) report to S3B and the SoCLabs contest.

7 Resourcing

Who	Role	h/wk
Lewis Watts	Lead;	5
Daniel Geng	Synthesis and place-and-route	10
Hee Chan Kim	Capacitor array, CDAC, analog library	10
Erick Gomez	Comparator, offset-trim arrays	12
Jaime Pryor	Bootstrap sampling switch	8
Benjamin Schwarz	Digital verification	5

Supervisor: Prof. Philip Leong. **External reviewers:** see Sec. 1.

Summary. Key infrastructure risks including synthesis, place-and-route, AMS co-simulation and extraction all working. Remaining tasks include analog block LVS, macro layout and integration. Our work will continue to be reviewed periodically by three external engineers.